

Technical Appendix

The Asset Side Channel: Artificial Intelligence and the Reallocation of Corporate Liquidity

Appendix A Technical Dictionary of AI Applications Generated Using BERT-base-Chinese Technology

Appendix A.1 BERT-base-Chinese Introduction

BERT (Bidirectional Encoder Representations from Transformers), a groundbreaking NLP technique proposed by Google researchers Jacob Devlin et al., was initially introduced in 2018. Its core innovation lies in bidirectional contextual representation learning through the pre-training and fine-tuning paradigm, which significantly advanced machines' language understanding capabilities. This architecture has been widely adopted for various NLP tasks including text classification, question answering, and named entity recognition.

BERT-base-Chinese is a variant of the BERT (Bidirectional Encoder Representations from Transformers) model, developed by Google, and is pre-trained specifically for the Chinese corpus. It inherits BERT's bidirectional and pre-training-fine-tuning paradigm, and aims to improve the performance of Chinese natural language processing (NLP) tasks, such as text classification, sentiment analysis, named entity recognition, and dictionary extension. The design of BERT-base-Chinese specifically takes into account the characteristics of the Chinese language (such as no clear word boundary and character level input) to make it perform well in Chinese scenes.

BERT-base-Chinese is based on Transformer encoder, consistent with standard BERT-Base, containing 12 Transformer encoders ($L=12$), 12 attention heads ($H=12$), a hidden dimension of 768 ($D=768$) and a total parameter of about 110 million. Compared to BERT-Base-Uncased (English), it is optimized for Chinese. Different from the English WordPiece word segmentation, BERT-base-Chinese directly takes a single Chinese character as the basic unit (Token), without the complex word segmentation preprocessing. For example, "I love artificial intelligence" (我爱人工智能) is broken down into "我", "爱", "人", "工", "智" and "能", retaining the semantic integrity of Chinese language. It uses large-scale Chinese corpora, such as Chinese Wikipedia and news corpus, to ensure that the model captures Chinese context features.

One prominent advantage of this approach is the dynamic nature of its character-level embeddings, which adapt to the surrounding context. Consider the characters '人', '工', '智', '能' in the word "人工智能" (artificial intelligence). These characters will receive different vector representations in the sentence "I love artificial intelligence" (我爱人工智

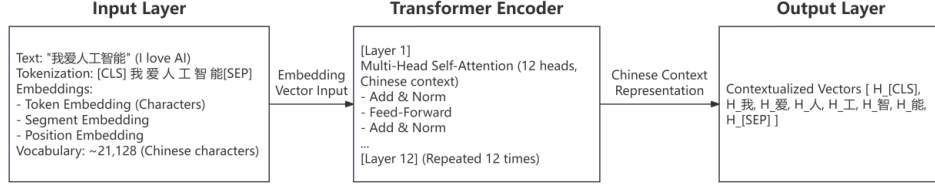


Fig. A1. A schematic diagram of BERT-base-Chinese

Notes: Figure A1 illustrates the architecture of the BERT-base-Chinese model used in our textual analysis. The process consists of three main stages: (1) the **Input Layer** tokenizes raw Chinese text and generates initial embeddings; (2) the **Transformer Encoder** uses 12 self-attention layers to capture deep contextual dependencies; and (3) the **Output Layer** produces contextualized vector representations for downstream tasks.

能) compared to “Artificial intelligence might cause unemployment to soar” (人工智能可能会导致失业率飙升). This ensures that the resulting word-level representation is also contextually sensitive.

Appendix A.2 Training of the BERT Model

We used Python to train the BERT-base-Chinese model, which was sourced from Hugging Face, following these steps:

Appendix A.2.1 Text Preprocessing

First, the annual reports of companies, crawled from the CNINFO website, were preprocessed by converting them from PDF to TXT format. We first fine-tune the BERT model with 20% of the annual reports and patent documents from the preceding year, so that the model can better adapt to the financial domain. To construct the training corpus, we randomly sample 20% of each year’s annual report texts. The sampled texts are then segmented into individual sentences using regular expressions based on common Chinese punctuation marks. The sentences are subsequently fed into the BERT-base-Chinese model for analysis.

Appendix A.2.2 Generating Average Word Embeddings with BERT

Following text preprocessing, we proceed with the generation of word vectors. First, sentences are processed using the BERT-base-Chinese model to compute contextual embedding vectors for each character (or subword unit) therein. Concurrently, the Jieba tokenization tool is employed to segment the original sentences into words, thereby defining word boundaries. Subsequently, by combining the character-level embedding vectors generated by BERT with the word boundaries identified by Jieba, a vector representation for each word in its specific context is obtained through an aggregation operation (e.g., averaging the character vectors constituting the same word).

To optimize computational efficiency, we only computed and retained vectors for words with a corpus frequency of 10 or greater, while simultaneously filtering out sentences exceeding 200 characters. Such lengthy sentences typically correspond to tables or footnotes in annual reports; processing them would not only increase computational overhead but could also introduce irrelevant noise vectors.

Appendix A.2.3 Dictionary Extension

(1) Based on the iResearch report China Artificial Intelligence Industry Research Report (2023), we selected eight seed terms (Machine Learning, Computer Vision, Knowledge Graph, Image Generation, Intelligent Speech, Natural Language Understanding, Large Language Model, Multimodal Large Model) as the foundation for expanding the artificial intelligence vocabulary. (2) We utilized word embeddings and cosine similarity to assess the similarity between candidate words and the seed terms. (3) We then removed duplicate words, and those unrelated to artificial intelligence, resulting in a final set of 45 words that constitute the AI lexicon.

Table A1: Vocabulary of Artificial Intelligence Technology

Key Words	Key Words
编程语言, Programming Language	知识库, Knowledge Base
大数据, Big Data	知识图谱, Knowledge Graph
大数据平台, Big Data Platform	知识型, Knowledge-based
大语言模型, Large Language Model	智能客服, Intelligent Customer Service
代码生成, Code Generation	智能算法, Intelligent Algorithm
动态数据, Dynamic Data	智能网, Intelligent Network
多模态大模型, Multimodal Large Model	智能音箱, Smart Speaker
分析模型, Analysis Model	智能语音, Smart Voice
计算机辅助, Computer-Aided	机器翻译, Machine Translation
计算机技术, Computer Technology	机器学习, Machine Learning (ML)
计算机控制, Computer Control	模式识别, Pattern Recognition
计算机领域, Computer Field	模型库, Model Library
计算机网络, Computer Network	数据模型, Data Model
计算机系统, Computer System	数学模型, Mathematical Model
计算机信息, Computer Information	数字模型, Digital Model
计算机行业, Computer Industry	图像识别, Image Recognition
人工智能, Artificial Intelligence	语音合成, Speech Synthesis
深度学习, Deep Learning	语音交互, Voice Interaction
数字图像, Digital Image	语音识别, Speech Recognition
图像处理, Image Processing	智能计算, Intelligent Computing
图像生成, Image Generation	自然语言, Natural Language
图像增强, Image Enhancement	自然语言理解, Natural Language Understanding
图形处理, Graphics Processing	计算机视觉, Computer Vision
移动机器人, Mobile Robot	自然语言处理, Natural Language Processing
知识化, Intellectualization	

Notes: This table presents the artificial intelligence (AI) technology dictionary, which is expanded using a fine-tuned BERT-base-Chinese model. By leveraging the model’s semantic learning capabilities on the corpora of corporate annual reports and patents, the dictionary identifies and expands upon core AI-related technological keywords.

Appendix B Machine Learning Models

We employed nine machine learning models to fit the control variables. The models and their formulations are described below.

Appendix B.1 Random Forest

Random Forest is an ensemble learning method proposed by Breiman (2001). It constructs multiple decision trees and averages their predictions. For regression predictions:

$$H(x) = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (1)$$

where $h_t(x)$ represents the predicted value of the t -th tree, and the final regression result $H(x)$ is the mean of all predicted trees.

Appendix B.2 XGBoost (Extreme Gradient Boosting)

XGBoost is an efficient implementation of Gradient Boosting Decision Trees (GBDT), proposed by Chen (2016). It enhances model performance through optimized objective functions.

$$y_i = \sum_{t=1}^T f_t(x_i) \quad (2)$$

Appendix B.3 Lasso

Lasso (Least Absolute Shrinkage and Selection Operator) incorporates an L1 regularization term into the objective function to achieve feature selection and model sparsity.

$$\beta = \operatorname{argmin}_{\beta} \left(\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j| \right) \quad (3)$$

Appendix B.4 Ridge Regression

Ridge Regression addresses multicollinearity by adding an L2 regularization term.

$$\beta = \operatorname{argmin}_{\beta} \left(\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2 \right) \quad (4)$$

Appendix B.5 KNN (K-Nearest Neighbors)

KNN predicts outcomes by calculating the distance between a test sample and training samples, selecting the K nearest neighbors.

$$y = \frac{1}{K} \sum_{i \in N_K(x)} y_i \quad (5)$$

Appendix B.6 Decision Tree

Decision Tree segments data into different regions through a series of conditional decisions.

$$y = \frac{1}{|R_m|} \sum_{i \in R_m} y_i \quad (6)$$

Appendix B.7 Support Vector Regression (SVR)

SVR is an extension of Support Vector Machines for regression.

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) K(x_i, x) + b \quad (7)$$

Appendix B.8 AdaBoost (Adaptive Boosting)

AdaBoost iteratively trains a series of weak learners, adjusting sample weights based on error rates.

$$y(x) = \sum_{t=1}^T w_t h_t(x) \quad (8)$$

Appendix B.9 Gradient Boosting

Gradient Boosting iteratively trains weak learners to reduce errors along the gradient descent direction.

$$y(x) = \sum_{t=1}^T \alpha_t h_t(x) \quad (9)$$

Table B1: Model Parameters

Model	Parameters
Linear Regression	-
Random Forest	n_estimators=100, max_depth=10, random_state=42
XGBoost	n_estimators=400, max_depth=50, random_state=42
Lasso	alpha=0.1, max_iter=10000
Ridge	alpha=0.1
KNN	n_neighbors=10
Decision Tree	max_depth=10, random_state=42
SVR	kernel='rbf', C=1.0, epsilon=0.1
AdaBoost	n_estimators=100, random_state=42
Gradient Boosting	n_estimators=100, max_depth=10, random_state=42

Notes: This table details the hyperparameter configurations for the machine learning algorithms employed in the Double Machine Learning (DML) framework. These algorithms are utilized to estimate the nuisance parameters in the first stage (i.e., predicting the treatment and outcome variables using the set of covariates). For each model, only the customized hyperparameters are listed; any parameters not explicitly reported herein are kept at their default values in the standard machine learning library (e.g., scikit-learn).

Appendix C Implementation Years of the AI Pilot-Zone Policy

This appendix reports the implementation years of the National New Generation Artificial Intelligence Innovation and Development Pilot-Zone policy. For Deqing County, we use the implementation year of its prefecture-level city, Huzhou, as a proxy.

Appendix D Construction of the Firm-Level EPU Index

To measure firm-level economic policy uncertainty (EPU), we fine-tune a BERT-base-Chinese model using text from Management Discussion and Analysis (MD&A) sections.

(1) We segment annual-report texts using Jieba and identify sentences containing both economic-policy-related and uncertainty-related vocabulary as candidate EPU-sensitive statements. The dictionary builds on Baker et al. (2016) and is adapted to the Chinese

Table C1: Implementation Years of the National New Generation Artificial Intelligence Innovation and Development Pilot-Zone Policy

City	Approval Year
Beijing	2019
Shanghai	2019
Tianjin	2019
Shenzhen	2019
Hangzhou	2019
Hefei	2019
Huzhou	2019
Chongqing	2020
Chengdu	2020
Xi'an	2020
Jinan	2020
Guangzhou	2020
Wuhan	2020
Suzhou	2021
Changsha	2021
Zhengzhou	2021
Shenyang	2021
Harbin	2021

Notes: Table C1 reports the official approval years for the establishment of the National New Generation Artificial Intelligence Innovation and Development Pilot Zones. For Deqing County, the implementation year of its prefecture-level city, Huzhou, is used as a proxy in the empirical analysis.

institutional context; Table D1 reports the screening vocabulary. Expert annotators identify 10,000 EPU-sensitive sentences, to which we add an equal-sized random sample of non-EPU-sensitive sentences.

(2) We split the resulting 20,000-sentence labelled dataset into training (60%), validation (20%), and test (20%) sets. We select fine-tuning parameters by grid search. The final classifier attains 99.23% accuracy on the held-out test set.

(3) We apply the trained classifier sentence by sentence to each MD&A section and define the firm-level EPU index as the proportion of sentences classified as EPU-related.

References

Scott R. Baker, Nicholas Bloom, and Steven J. Davis. Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636, 2016.

Table D1: Vocabulary of Economic Policy and Uncertainty

Economic-policy terms	Uncertainty terms
市政, Municipal administration	风险, Risk
政策, Policy	经营风险, Operational risk
货币政策, Monetary policy	市场风险, Market risk
政策鼓励, Policy encouragement	信用风险, Credit risk
国家, State	不确定性, Uncertainty
扩内需, Expand domestic demand	不确定, Uncertainty
保增长, Ensure economic growth	波动, Fluctuation
促发展, Promote development	变化, Change
产业发展, Industrial development	改变, Change
法律, Law	徘徊, Fluctuate
法规, Regulations	不稳, Unstable
行业政策, Industry policy	不稳定性, Instability
产业政策, Industrial policy	不稳定, Unstable
宏观政策, Macro policy	不寻常, Unusual
国民经济, National economy	错综复杂, Complex
有关部门, Relevant departments	非常复杂, Very complicated
产业结构调整, Industrial structure adjustment	纷繁复杂, Multifarious
产业结构, Industrial structure	纷纭复杂, Confusing and complicated
当地政府, Local government	十分复杂, Extremely complicated
政府, Government	结构复杂, Complex structure
经济政策, Economic policy	变得复杂, Become complicated
政策走势, Policy trend	风云变幻, Dramatic changes
所得税, Income tax	风云突变, Sudden dramatic change
增值税, Value-added tax	矛盾突出, Prominent contradictions
税收减免, Tax reduction and exemption	突变, Sudden change
税收优惠, Tax preference	复杂, Complex
刺激政策, Stimulus policy	复杂多变, Complex and changing
信贷令, Credit restriction order	诡谲多变, Devious and volatile
限购令, Purchase restriction order	阵痛, Pangs
保障房, Affordable housing	过渡, Transition
宏观调控, Macro-control	问责, Accountability
证监会, Securities regulatory commission	整顿, Rectification
国家政策, National policy	危险, Danger
政治, Politics	动荡, Turmoil
军事, Military	动荡不安, Turmoil and instability
政策环境, Policy environment	动荡不定, Unpredictable instability
宏观, Macro	多变性, Variability
政府补助政策, Government subsidy policy	振荡下行, Volatile downward trend
调控政策, Regulation policy	震荡, Volatility
政府补助, Government subsidy	震荡不安, Volatility and instability
税收政策, Tax policy	政治波动, Political fluctuation
政策扶持, Policy support	难以确定, Difficult to determine
减税降费, Tax and fee reductions	难以预测, Hard to predict
转移支付, Fiscal transfer payments	难以预料, Hard to anticipate
试点政策, Pilot policy	难以捉摸, Elusive
经济, Economics	接受考验, Be tested
	混乱, Chaos

Notes: Table D1 presents the dictionary used for preliminary screening. It builds on Baker et al. (2016) and is supplemented with vocabulary tailored to the Chinese institutional context.